

Exp No: 11

STUDY OF GPIO & BIT ADDRESSING IN AT89C51

Program:

```
#include <reg51.h>

sbit LED1 = P1^0;
sbit LED2 = P1^1;

void delay()
{
    unsigned int i,j;
    for(i=0;i<200;i++)
    for(j=0;j<1000;j++);
}

void main()
{
    while(1)
    {
        LED1 = 1;
        LED2 = 0;
        delay();
        LED1 = 0;
        LED2 = 1;
        delay();
    }
}
```

Output

C:\Users\yada\OneDrive\Documents\embedded\EXP 111.uvproj - µVision

File Edit View Project Flash Debug Peripherals Tools SVCS Window Help

Registers

Register	Value
r0	0x00
r1	0x00
r2	0x00
r3	0x00
r4	0x00
r5	0x00
r6	0x00
r7	0x00
Sys	0x00
a	0x00
b	0x00
c	0x07
sp_max	0x07
dptr	0x0000
PC	C:0x081D
status	329
sec	0.000422s
psw	0x00

Disassembly

```
10: void main()
11: {
12:   while(1)
13:   {
14:     LED1 = 1;
15:     LED2 = 0;
16:     delay();
17:     LED1 = 0;
18:     LED2 = 1;
19:     delay();
20:   }
21: }
```

EXP 111.LC

```
1 #include <reg51.h>
2 sbit LED1 = P1^0;
3 sbit LED2 = P1^1;
4 void delay()
5 {
6   unsigned int i,j;
7   for(i=0;i<200;i++)
8   {
9     for(j=0;j<1000;j++)
10    {
11      void main()
12      {
13        while(1)
14        {
15          LED1 = 1;
16          LED2 = 0;
17          delay();
18          LED1 = 0;
19          LED2 = 1;
20          delay();
21        }
22      }
23    }
24  }
25 }
```

Parallel Port 1

Port 1	7	6	5	4	3	2	1	0
P1: 0x00	☒	☒	☒	☒	☒	☒	☒	☒
Pins: 0x00	☒	☒	☒	☒	☒	☒	☒	☒

Command

Running with Code Size Limit: 2K

Load "C:\Users\yada\OneDrive\Documents\embedded\Objects\EXP 111"

Call Stack - Locals

Name	Location/Value	Type
MAIN	C:0x081D	

ASM ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE COVTOFILE

Simulation

11: 30.36388346 sec L10 C:13 CAP NUM SCRL OVR: B/W

33°C Partly sunny

Search

ENG IN 11:06 27-07-2026